MATH 420-01

QUESTION 2 on this weeks homework: When you are trying to show two sets are equal, say $R = S$, sometimes it works to show $R \subseteq S$ and $S \subseteq R$.

QUESTION 4: Examples of vector spaces: $\mathbb{R}^n$, $\mathbb{R}$, $\mathbb{C}$, $\{0\}$, Matrices, $\mathbb{P}_n$

MIDTERM: 8.5 by 11 one sided formula sheet will be allowed. There will be "choice" of problem on the exam. Choose 4 from 6.

2.4 Invertibility and Isomorphisms

Recall:

$I_V : V \to V$ is the linear transformation such that $I_V(v) = v, \ \forall\, v \in V$.

Example:

Find $[I_V]_\alpha^\beta$ when $\alpha = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$ and $\beta = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

apply transformation

Example:

Find $[I_V]_\alpha^\beta$ when $\alpha = \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$ and $\beta = \left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \right\}$

$$\begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 1 \\ 2 & 0 & 1 & 1 & -1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}$$

we need lin comb's of β vectors

In[ ]:= RowReduce$\left[\begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 1 \\ 2 & 0 & 1 & 1 & -1 & 1 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{bmatrix}\right]$

Out[ ]= $\begin{pmatrix} 1 & 0 & 0 & 1 & -1 & 0 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 0 & 0 & 1 & -1 & 1 & 1 \end{pmatrix}$

We could have solved three systems:

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 2 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 0 & 1 \\ 2 & 0 & 1 & -1 \\ 1 & 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 & 0 & 1 \\ 2 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$$

$$[I_V]_\alpha^\beta = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 2 & 1 \\ -1 & 1 & 1 \end{pmatrix}$$

$$[I_V]_\beta^\beta = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Definition:

Let V and W be vector spaces over a field $\mathbb{F}$. Let $T : V \to W$ be a linear transformation. A transformation $S : W \to V$ is called the *inverse* of T if $S\,T = I_V$ and $T\,S = I_W$. If T has an inverse, it is said to be *invertible*, and the inverse is denoted by T^{-1}.

Invertible and nonsingular are synonyms

Not Invertible and singular are synonyms.

Something is invertible or it is NOT.

Theorem:

Let V and W be vector spaces over a field $\mathbb{F}$. Let $T : V \to W$ be an invertible linear transformation. Then

(i) $T^{-1} : W \to V$ is a linear transformation.

(ii) $\left(T^{-1}\right)^{-1} = T$.

(iii) If V is finite dimensional, then $\dim(W) = \dim(V)$.

(iv) If $S : V \to W$ is an invertible linear transformation, then $(S\,T)^{-1} = T^{-1}\,S^{-1}$.

(v) T is one-to-one and onto.

PROOF: